

PRODUCT PACKET

Decision Bajo Presion - Week 6 Build Packet

A measurable rehearsal for decisions under unexpected emergency conditions.

Submission detail	Value
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Coursework	Week 6 individual working slice
Product	Decision Bajo Presion
Live URL	https://decision-bajo-presion-week-6.vercel.app
GitHub	https://github.com/davidbuzali/decision-bajo-presion-week-6
Deployment	Deployment 2 - verified Ready
Data boundary	Synthetic demonstration records only

Prepared for final Week 6 submission.

Decisión Bajo Presión - Week 6 Product Packet

Declared vacuum

The MEASUREMENT vacuum - zero data on whether drills change behavior.

The working slice measures whether one narrowly defined decision behavior transfers from a baseline rehearsal to a different, unseen scenario. Compliance-Upgrade remains a possible route to market through existing civil-protection budgets, but it is not the primary vacuum. The Family and Insurance-Bridge vacuums are outside this week's slice.

Problem in my words

Emergency plans are usually taught as a fixed sequence: hear the alert, follow the route, reach the assembly point. The failure begins when that sequence stops working. A blocked exit, missing person, failed message, or conflicting sign forces an adult to interpret uncertainty and choose an action, yet ordinary compliance records mainly prove attendance or completion.

This slice attacks **The MEASUREMENT vacuum**: there is almost no useful evidence showing whether emergency drills change behavior. The team's specific interpretation is behavior measurement. The product creates a short, repeatable rehearsal that records narrow observable decisions, supports a human debrief, and then changes the situation for an unseen retest. It does not claim that a participant, school, or organization is prepared. For critical behaviors, the digital result remains pending until a human records a physical transfer test.

Exact user

Primary user: A teacher or school-brigade member at a private school in Mexico City that already maintains a formal civil-protection program with a registered consultant.

The user is comfortable with a browser but is not a gamer. They have limited time between school responsibilities, may be using shared equipment, and need clear Spanish instructions. They must be able to pause without penalty and use mouse, keyboard, touch, or voice. A non-immersive alternative must remain available.

Secondary actor: The school's civil-protection consultant or safety lead, who facilitates the debrief, reviews aggregate evidence, and decides what must be tested physically.

Success definition

Before the module closes, the adult user can:

1. Complete a three-decision baseline scenario in a browser-based 3D environment.
2. Review an event trace with a human facilitator that describes actions without assigning psychological traits or a generic preparedness score.
3. Face a different, unseen scenario selected from the baseline evidence and make the corresponding decisions by clicking or speaking.
4. See whether a previously missed narrow behavior was demonstrated in the retest.

5. See every critical result labeled **Pending physical validation**, with no digital control capable of declaring competence.

The slice succeeds technically when every decision is timestamped, the adaptive retest never repeats the exact baseline disruption, voice and click inputs produce the same decision event, the pause control works at every decision, and the final view clearly separates digital evidence from the human training decision.

The working slice - screen by screen

The slice is one short adult rehearsal lasting approximately six to eight minutes. It contains five screens and no account creation.

Screen 1 - Before you begin

The user sees three limits before entering the simulation:

- This is a **screen-based 3D rehearsal**, not virtual reality and not a prediction of real-world safety.
- The module records predefined decisions and relative timestamps, not identity, emotion, courage, panic, or biometric information.
- The user can pause or leave without penalty and can complete the full module without voice input.

The user selects normal or reduced motion and may explicitly enable bounded Spanish voice commands. Voice permission is not requested until the user chooses it.

Screen 2 - Baseline scenario A

The 3D scene shows a generic Mexican school corridor. It is not a digital twin of a real building. The adult encounters three decision points:

1. **The alert begins:** observe the environment and follow the school's current instruction, move immediately without checking conditions, or continue the previous activity.
2. **The primary exit is blocked:** verify and use the marked alternate route, attempt the obstructed route, or wait without communicating.
3. **An assigned person is not accounted for:** report the missing status and request coordinated support, leave the group to search alone, or close the count without reporting the discrepancy.

Each choice changes the visible 3D state and records only an allow-listed action code, scenario version, relative timestamp, and input mode. The simulation does not call these choices proof of competence. Scenario details and final option wording must be validated by a Mexican civil-protection professional before use outside the classroom prototype.

Screen 3 - Observable trace and human debrief

The user and human facilitator see a chronological trace such as:

- 00:18 - Observed the route sign
- 00:31 - Selected the blocked primary exit
- 00:47 - Reported an accountability discrepancy

The adaptive system highlights one narrow behavior for the retest, such as `checks_route_status`, but it does not assign a preparedness score or reveal the correct response to the next scenario. The user must confirm that a human debrief occurred before continuing.

Screen 4 - Unseen scenario B

The system selects a different disruption based on the missed or weakest observable behavior from scenario A:

Baseline evidence	Different retest disruption	Behavior tested again
Attempted a blocked exit	The normal alternate stairwell is closed and a temporary route sign appears	<code>checks_route_status</code>
Lost accountability	A participant is missing after the group changes assembly area	<code>maintains_accountability</code>
Abandoned or ignored an assistance assignment	The assigned person cannot use the newly selected route	<code>requests_assistance</code>
No target behavior missed	Two signs conflict after a route change	<code>verifies_information</code>

The user again chooses by button, keyboard, or an allow-listed voice command. Scenario B never repeats scenario A's exact layout or disruption.

Screen 5 - Comparison and next action

The final screen reports only one evidence statement for the targeted behavior:

- **Demonstrated in both simulations**
- **Demonstrated after the debrief in the unseen retest**
- **Not yet demonstrated - requires more rehearsal**

For every critical behavior, the screen also shows **Pending physical validation**. It directs the human safety lead to test the behavior during the school's physical micro-drill. There is no button that marks the participant competent, the school prepared, or the issue closed.

Bounded voice grammar

Voice is an optional control method, not an open-ended assistant. The first build recognizes only a small Spanish grammar mapped to the same action codes as the buttons:

- `ruta alterna`
- `reportar persona`
- `pedir apoyo`
- `esperar instrucción`
- `pausar`

Unrecognized speech is discarded immediately and produces a neutral message asking the user to select a visible option. Raw audio and transcripts are never stored.

Image-generated mockup

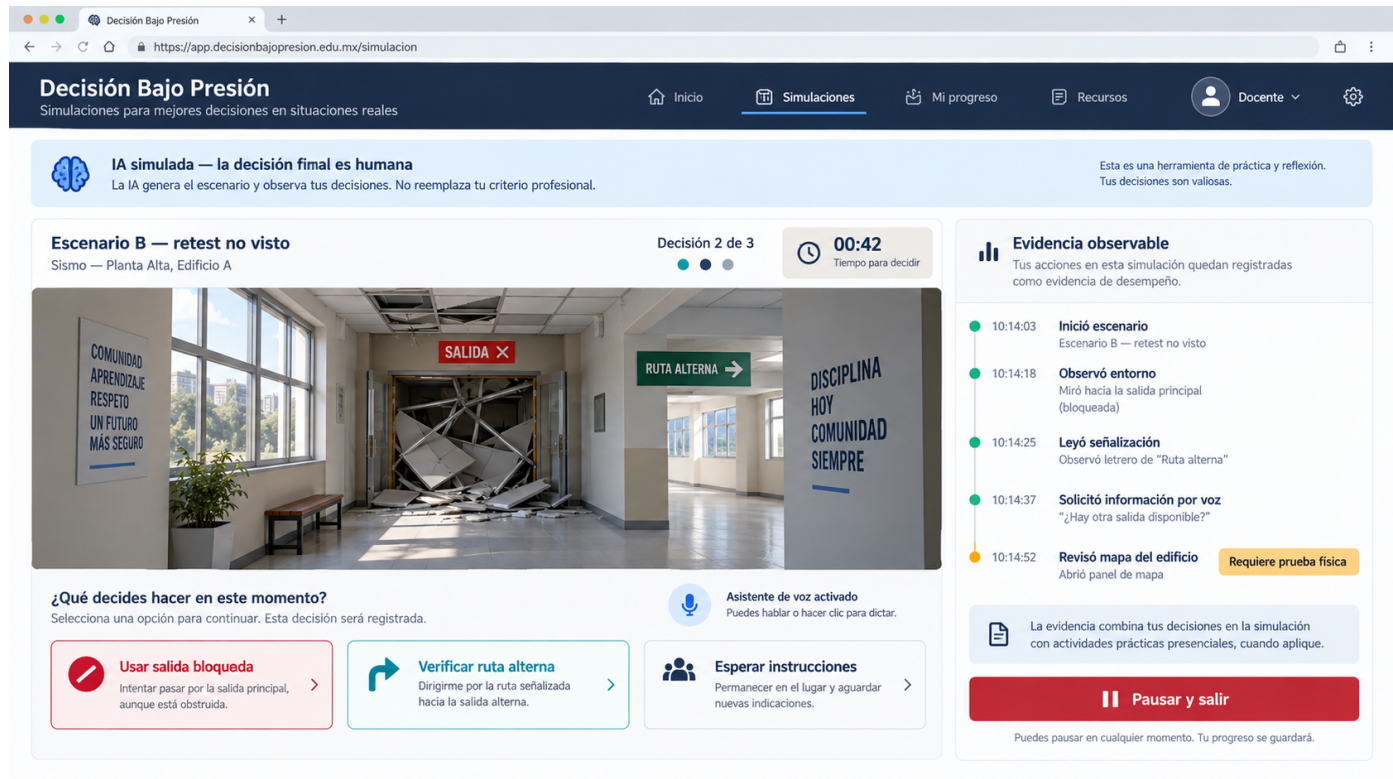


Image-generated mockup of the adult retest and observable evidence trace

The mockup represents **Screen 4 - Unseen escenario B**. It was generated with an AI image tool and represents the target interaction, not the final implementation. It deliberately includes the labels **Escenario B - retest no visto**, **IA simulada - la decisión final es humana**, **Requiere prueba física**, and **Pausar y salir**. The final coded interface may simplify the visual scene, but it must preserve this information hierarchy, the neutral tone, and the permanent exit control.

Feature flow

Mermaid source

```
graph TD
    A[Screen 1: limits and access mode] --> B{Enable optional voice?}
    B -->|No| C[Screen 2: baseline scenario A]
    B -->|Yes| D[Request browser microphone permission]
    D --> C
    C --> E{Baseline decision point}
    E -->|Click or keyboard| F[Validate allow-listed action]
    E -->|Bounded voice phrase| F
    F --> G[Record action code + relative timestamp + input mode]
    G --> H{Three decisions complete?}
    H -->|No| E
    H -->|Yes| I[Screen 3: observable trace]
    I --> J[Human-facilitated debrief]
    J --> K[Adaptive logic selects a different disruption]
    K --> L[Screen 4: unseen scenario B]
    L --> M[Record retest actions through the same validator]
    M --> N[Compare one narrow behavior]
    N --> O[Screen 5: evidence statement]
    O --> P[Always show pending physical validation for critical behavior]
    P --> Q[Human safety lead chooses physical micro-drill action]
    Q -- "Outside this working slice" --> R[Corrected behavior closes issue; otherwise it stays open]
```

Actor swimlane

Mermaid source

```

flowchart LR
    subgraph U[Adult teacher or brigade member]
        direction TB
        U1[Chooses access mode]
        U2[Makes three baseline decisions]
        U3[Reflects during debrief]
        U4[Completes unseen retest]
        U5[Reviews evidence and next action]
    end

    subgraph V[Optional voice layer]
        direction TB
        V1[Recognizes bounded Spanish phrase]
        V2[Maps phrase to allow-listed action code]
        V3[Discards unrecognized speech]
    end

    subgraph S[Browser simulation]
        direction TB
        S1[Presents generic 3D baseline]
        S2[Validates and logs baseline actions]
        S3[Shows event trace]
        S4[Presents different 3D retest]
        S5[Validates and logs retest actions]
        S6[Shows comparison and physical-validation gate]
    end

    subgraph A[Transparent adaptive logic]
        direction TB
        A1[Reads allow-listed baseline events]
        A2[Selects one behavior target]
        A3[Chooses a different disruption]
        A4[Compares the targeted behavior]
    end

    subgraph H[Human safety lead]
        direction TB
        H1[Reviews factual event trace]
        H2[Conducts debrief]
        H3[Chooses physical micro-drill action]
        H4[Later closes issue or keeps it open]
    end

    U1 --> S1
    S1 --> U2
    U2 -->|Click or keyboard| S2
    U2 -->|Optional speech| V1
    V1 --> V2
    V1 -->|No allow-listed match| V3
    V2 -->|During baseline| S2
    S2 --> A1
    A1 --> A2
    S2 --> S3
    S3 --> H1
    H1 --> H2
    H2 --> U3
    U3 --> A3
    A2 --> A3
    A3 --> S4
    S4 --> U4
    U4 -->|Click or keyboard| S5
    U4 -->|Optional speech| V1
    V2 -->|During retest| S5
    S5 --> A4
    A4 --> S6
    S6 --> U5
    S6 --> H3
    H3 -. Physical drill outside slice .-> H4

```

Benchmark

The best existing solution on Earth for this is: FLAIM FTS with FLAIM Capture, which connects immersive emergency scenarios to performance recording, analytics, and data-informed after-action review for professional responders.

Mine differs or localizes by: Decisión Bajo Presión brings the rehearse-record-debrief loop into the existing CDMX school civil-protection program and Responsable Oficial workflow, begins with an accessible browser simulation, tests transfer through a different unseen scenario, and keeps critical findings open until a human-led physical drill validates the behavior.

Long view - three-year light charter

In three years, Decisión Bajo Presión becomes a consultant-operated rehearsal library for Mexican schools and workplaces, with locally validated scenarios, accessible interaction modes, and comparable evidence across repeated training cycles. Organizations use it to see whether narrow decision behaviors transfer across unseen simulations and physical drills, while unresolved failures flow to a responsible owner, corrective action, and closure test. It remains human-led training infrastructure-not a fear engine, surveillance product, or automatic certification of safety.

Scope cut

This week I am deliberately **not** building:

- **Headset VR or WebXR:** the required simulation layer is one lightweight browser-based 3D corridor, explicitly labeled as a screen-based rehearsal.
- **A digital twin or certified route map:** the scene is generic and cannot validate a real building, evacuation route, or regulatory plan.
- **A trained predictive model:** the “AI” layer is transparent adaptive logic that chooses an unseen retest from allow-listed behavior evidence; it does not predict real-world performance.
- **A backend or institutional platform:** there are no accounts, authentication, database, dashboards across schools, permanent profiles, or cloud-stored voice transcripts.
- **Student, family, insurer, or parent workflows:** the only participant is an adult teacher or brigade member, and the only secondary actor is a human safety lead.
- **Realistic disaster physics:** there is no crowd simulation, structural model, earthquake model, biometric stress measurement, gaze tracking, emotion recognition, or fear-adaptive intensity.
- **Open-ended conversational AI:** voice recognizes only the small command grammar listed above and discards anything it cannot map safely.
- **Compliance or competence certification:** the prototype cannot approve a civil-protection program, replace a required drill, close a critical finding, or declare a person or school prepared.
- **The physical drill itself:** the final screen creates the human next action and preserves the open issue; actual transfer testing and closure happen outside this working slice.

Architecture and stack

The working slice is a client-only single-page application. It has no backend, account system, database, analytics service, or external AI API. [Vite's static-deployment guidance](#) supports direct Git-based deployment to Vercel, where the default production build is generated in `dist`.

Module	Free technology	Contract and responsibility
Application shell	React + TypeScript + Vite	Owns the explicit session phases: intro, baseline, debrief, retest, comparison, and paused; renders Spanish UI and accessibility controls
3D scene	Three.js through React Three Fiber	Receives a typed <code>ScenarioDefinition</code> , renders only lightweight geometry, signs, and route state, and emits a selected option; it contains no scoring logic
Non-3D equivalent	Semantic React controls	Presents the same decision options and produces the same action codes when WebGL is unavailable, reduced motion is selected, or the user prefers a flat view
Action validator	Pure TypeScript module	Accepts only known <code>DecisionId</code> , <code>ActionCode</code> , and <code>InputMode</code> values; click, keyboard, and recognized speech must all pass through this one validator
In-memory event log	Pure TypeScript reducer	Appends only <code>scenarioId</code> , <code>decisionId</code> , <code>actionCode</code> , <code>relativeTimeMs</code> , and <code>inputMode</code> ; clears on reload and never stores names, raw speech, free text, or psychological labels
Adaptive selector	Pure deterministic TypeScript function	Reads baseline action codes, chooses one behavior target, and returns a different allow-listed retest scenario; it never predicts real-world performance
Voice adapter	Optional Web Speech API SpeechRecognition adapter	Converts a recognized phrase into an allow-listed action code, then immediately discards the phrase; failure or unsupported browsers fall back to visible controls
Evidence comparison	Pure TypeScript function	Produces one of the three approved evidence statements plus an immutable physical-validation requirement for critical behaviors
Test harness	Vitest + React Testing Library + Playwright	Tests domain logic, interaction parity, accessibility paths, and the full browser flow
Deployment	GitHub + Vercel static deployment	Creates a first public deployment before testing and a second verified deployment after the documented bug fix

Core data contracts

ts

```
type SessionPhase =
  | "intro"
  | "baseline"
  | "debrief"
  | "retest"
  | "comparison"
  | "paused";

type InputMode = "pointer" | "keyboard" | "voice";

type DecisionEvent = {
  scenarioId: string;
  decisionId: string;
  actionCode: string;
  relativeTimeMs: number;
  inputMode: InputMode;
};

type ComparisonResult = {
  behaviorCode: string;
  evidenceStatement:
    | "demonstrated_both"
    | "demonstrated_after_debrief"
    | "not_yet_demonstrated";
  physicalValidation: "pending";
};
```

Names, emails, school identifiers, raw audio, transcripts, absolute timestamps, location, biometrics, emotion labels, and readiness scores are forbidden fields.

Multiplicative Dragon Stack

The stack is **simulation/3D + adaptive logic + voice**. It is multiplicative because the three layers alter one shared rehearsal-and-evidence loop rather than appearing as unrelated demonstrations.

Mermaid source

```
graph LR
  A["A[3D baseline presents disruption]"] --> B["B[Shared action validator]"]
  B --> V["V[Optional voice phrase] -->|Mapped to same ActionCode as a button| B"]
  V --> K["K[Click or keyboard] --> B"]
  K --> B
  B --> C["C[Typed observable event log]"]
  C --> D["D[Adaptive selector chooses one missed behavior]"]
  D --> E["E[Different retest ScenarioDefinition]"]
  E --> F["F[3D scene changes route, sign, or assistance condition]"]
  F --> B
  B --> G["G[Human-readable evidence comparison]"]
  G --> H["H[Human debrief and physical-drill decision]"]
```

- **3D is load-bearing:** it makes a route obstruction or changed assistance condition visible and stateful; removing it eliminates the simulated disruption.
- **Adaptive logic is load-bearing:** baseline evidence selects which different scenario definition the 3D scene renders next; removing it turns the retest into a fixed quiz.
- **Voice is connected, not decorative:** a recognized bounded command enters the exact validator and event schema used by buttons, so it can influence both the visible scene and the adaptive retest.
- **Human judgment remains load-bearing:** the software can compare narrow evidence, but only the safety lead chooses training action and only a physical drill can validate a critical behavior.

Adaptive policy

The product uses an interpretable adaptive policy rather than claiming a trained predictive model. Each baseline choice maps to a narrow behavior code such as `checks_route_status`, `maintains_accountability`, or `requests_assistance`. The selector chooses an unseen scenario whose surface details differ but whose decision requires the missed behavior. If no target behavior was missed, it selects a deterministic alternate scenario that introduces conflicting information and tests whether the successful behavior transfers.

The comparison reports only one of three statements:

- **Demonstrated in both simulations**
- **Demonstrated after debrief in the unseen retest**
- **Not yet demonstrated - requires more rehearsal**

All critical behaviors additionally display **Pending physical validation**. These labels are simulated decision support for a human facilitator, not competence judgments.

Security and ethics floor

Required check	Design decision	Verification before deployment
1. No secrets in code or repository	The app uses no API, AI, database, analytics, or authentication keys; <code>.env*</code> is ignored except <code>.env.example</code>	Search tracked files for common key patterns and inspect <code>git diff --cached</code> before every push
2. Authentication if personal data is stored	No personal data is requested or stored, so authentication is intentionally unnecessary	Confirm there are no name, email, school, free-text, upload, or profile fields and that reload clears the session
3. Row Level Security for user tables	Not applicable because the slice has no Supabase project, database, or user table	Confirm the dependency list and network panel contain no database client or data-write request
4. Validate every input	All decisions and voice results must match typed allow lists; numeric durations are generated internally and clamped; there is no free-text input	Unit-test every accepted command and representative rejected, empty, overlong, and unrelated phrases
5. Invented demo data only	The generic scene, session IDs, people, and organization references are fictional and labeled as demonstration content	Review every fixture and screenshot before the submission is exported

Voice is disabled by default and never blocks completion. `SpeechRecognition` has limited browser availability, and some implementations may send audio to a browser provider's recognition service; the start screen must disclose this before permission is requested. The application itself stores neither audio nor recognized transcripts: it retains only the resulting allow-listed action code and marks its input mode as `voice`.

The interface labels every adaptive result **IA simulada - la decisión final es humana**. Users can pause or exit without penalty, and no scenario contains personalized trauma, recognizable victims, cloned voices, graphic harm, hidden manipulation, or fear-adaptive intensity.

Test plan

Mechanical pass

ID	Level	Test	Passing evidence
M01	Unit	Every baseline behavior mapping selects the intended retest family	Adaptive-selector tests pass for all mappings
M02	Unit	Retest scenario ID and disruption differ from baseline	No returned pair duplicates both fields
M03	Unit	Click, keyboard, and recognized voice phrases resolve through one allow list	Equal action codes are produced for equivalent inputs
M04	Unit	Unknown, empty, unrelated, and overlong speech is rejected	No event is appended and neutral retry copy is returned
M05	Unit	Comparison uses only the three approved statements	Exhaustive result test passes
M06	Unit	Every critical result keeps <code>physicalValidation: "pending"</code>	No function or UI action can create another value
M07	Unit	Event records contain only five allowed fields	Exact-key assertion passes; forbidden-field search returns zero
M08	Component	Voice permission is requested only after explicit enablement	Permission mock has zero calls before the enable action
M09	Component	Voice failure, denial, or unsupported browser preserves visible controls	The same scenario remains completable by keyboard
M10	Component	Pause is available at every decision and creates no failure event	Phase becomes paused; evidence log remains unchanged
M11	Accessibility	The entire flow works with keyboard and the flat-view alternative	Focus order and action parity assertions pass
M12	End-to-end	Baseline → trace → human-debrief acknowledgment → different retest → comparison	Playwright completes the journey and finds the physical-validation label
M13	Privacy	Reload removes all session evidence	Initial state returns with an empty event log
M14	Responsive	Core task works at desktop and narrow mobile widths without clipped decisions	Automated screenshots and manual inspection pass
M15	Build/security	Tests, production build, secret scan, and invented-fixture review pass	Commands and results are recorded in <code>docs/TESTING.md</code>

After M01-M15 pass, make **Deployment 1** and manually repeat the primary journey at the public URL. Record at least one real defect discovered during the mechanical pass, including reproduction steps, expected behavior, actual behavior, root cause, and screenshot; fix it, rerun every affected test plus M12 and M15, then make and verify **Deployment 2**. Never invent the required bug or label an unverified deployment as successful.

Persona pass

Synthetic user: **Laura Hernández, 47, a primary-school teacher and brigade volunteer in CDMX.** She uses WhatsApp and school platforms but has never played a 3D game, reads instructions carefully, dislikes being judged by software, and may abandon a tool if she cannot tell whether a button records a final answer. She has mild motion sensitivity and prefers reduced motion plus ordinary buttons; voice is optional, not assumed to be more accessible.

In a fresh conversation, provide Laura's persona and show screenshots of Screens 1-5 in order. Ask her to attempt the task in character while narrating what she thinks each screen means, what she would select, where she hesitates, and where she would quit; do not tell her the intended interpretation until the pass is complete.

Log each observation in docs/PERSONA.md with this structure:

Screen	Attempted action	Confusion or hesitation	Severity	Evidence	Fix decision
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Rank issues as blocks completion, changes the meaning, or slows/confuses. Fix the highest-severity issue before the deadline, capture the changed screen, and repeat that screen with the same persona instructions to determine whether the confusion was reduced. The pass must specifically test whether Laura notices **Pausar y salir**, can switch away from 3D motion, understands that a decision is recorded when selected, and distinguishes simulated improvement from pending physical validation.

Blueprint conditions traceability

Blueprint condition	How this slice honors it
Narrow observable actions only	The log stores allow-listed decisions and timestamps, never psychological or survival scores
Human debrief + physical transfer	A facilitated debrief precedes the retest; critical results remain pending a physical micro-drill
Start operationally small	One adult, two short browser scenarios, three decisions each, no headset required
Fit Mexican civil protection	The secondary actor is an existing consultant or safety lead, not an automated replacement
Shadow clause	No personalized trauma, cloned voices, recognizable victims, fear adaptation, or permanent named profile
Failures remain visible	An unresolved critical behavior cannot be converted into completion or competence by the digital module